NOVA-RF1

Original firmware-compatible hand-held RF / NFC / sub-GHz multi-tool

NOVA-RF1 is an **original open-hardware** hand-held tool for RFID/NFC, 125 kHz, 13.56 MHz, sub-GHz (300-928 MHz), infrared, 1-Wire/iButton and BLE work. It is built around the same silicon class as popular multi-tools and is designed to run the open-source Flipper Zero firmware (GPLv3) and its community forks — but it is **not** a Flipper Zero: its schematic, PCB, outline, silkscreen and name are its own, and it uses none of Flipper Devices' trademarks, logo, dolphin mascot or design files.

Board 100 × 60 mm, 4-layer, two-sided SMT, in a two-part printed enclosure. Radios require type approval (FCC / CE / WPC-ETA+BIS) before sale; transmit only on bands and powers you are licensed for.

1. How it works — working principle

A single **STM32WB55** (Cortex-M4 application core + Cortex-M0+ BLE stack) is the brain. It talks to each radio / peripheral over two SPI buses and a handful of GPIO/timer lines, exactly as the firmware's pin map declares. Power comes from USB-C or a 1-cell Li-Po through a charger and switching regulators. Each subsystem:

MCU + Bluetooth LE — STM32WB55RGV6

The application core runs the UI, protocol stacks and app logic; the on-chip M0+ runs BLE 5. A 32 MHz crystal (placed directly under the chip) clocks the radio; a 32.768 kHz crystal keeps the RTC. The internal SMPS (with its 10 uH inductor) powers the radio efficiently. The 2.4 GHz BLE signal leaves through a band-pass filter and pi-match to a chip antenna.

Sub-GHz radio — CC1101 (300-928 MHz)

On the SPI-R bus. The MCU configures frequency, modulation and power; the CC1101's differential RF port goes through a balun + LC match to two SPDT RF switches that select band/antenna, then to a helical/whip antenna via a U.FL connector. Used to capture and replay 433/868/915 MHz remotes.

13.56 MHz NFC — ST25R3916

Also on SPI-R. A 27.12 MHz crystal clocks it; its differential driver feeds an EMC + matching network (L1/L2 270 nH + caps) into a PCB loop antenna, and the receiver reads card modulation on RFI_P/RFI_N. Reads/writes/emulates ISO14443 A/B, ISO15693 and FeliCa cards.

125 kHz RFID — discrete analog front-end

Not a single chip: an H-bridge of FETs drives the read coil with a 125 kHz carrier gated by the MCU timer; the returning amplitude-modulated signal is band-pass amplified by two op-amps, sliced by a comparator against a 1.65 V mid-rail, and read back as digital data. Reads EM4100/HID-class tags.

Infrared

An IR LED bank (driven by a FET from a timer pin) transmits 38 kHz-modulated remote codes; a TSOP demodulator receives and decodes incoming IR for learning remotes.

1-Wire / iButton

An open-drain GPIO with ESD protection reads and emulates Dallas/iButton (DS1990-class) keys on a spring-contact pad.

Display, storage, UI

A 128×64 monochrome LCD (ST7565R, SPI-D bus) shows the UI; a microSD card (also SPI-D) holds apps and captured data; a 5-way + back button cluster drives navigation; an RGB LED (LP5562 driver), a speaker and a vibromotor give feedback.

Power path

USB-C 5 V feeds a BQ25896 1-cell Li-ion charger with 5 V OTG boost; a BQ27220 fuel gauge tracks charge over the I2C power bus. Two LM3281 buck converters make the 3.3 V core and peripheral rails; an RS3221 LDO gives a clean analog rail; a load switch cuts the peripheral rail in deep sleep for long standby. The battery pack is external, wired to the X10 connector.

Expansion

An 18-pin GPIO header exposes 8 user GPIO, both SPI buses, UART, the I2C power bus, SWD and 3V3/5V/GND — e.g. for an ESP32-S3 Wi-Fi/BLE add-on. A Tag-Connect pad programs the MCU over SWD.

Signal flow (block diagram)

USB-C / Li-Po → BQ25896 charger + LM3281 bucks → 3.3 V rails → **STM32WB55**

- SPI-R → CC1101 (sub-GHz) → balun → RF switch → U.FL antenna
- SPI-R → ST25R3916 (NFC) → matching net → 13.56 MHz loop
- timers/GPIO → 125 kHz coil driver + op-amp/comparator RX
- SPI-D → ST7565 display + microSD
- GPIO → IR TX/RX, iButton, buttons, RGB LED, speaker, vibro
- on-chip 2.4 GHz → BLE filter → chip antenna

2. Bill of materials & cost

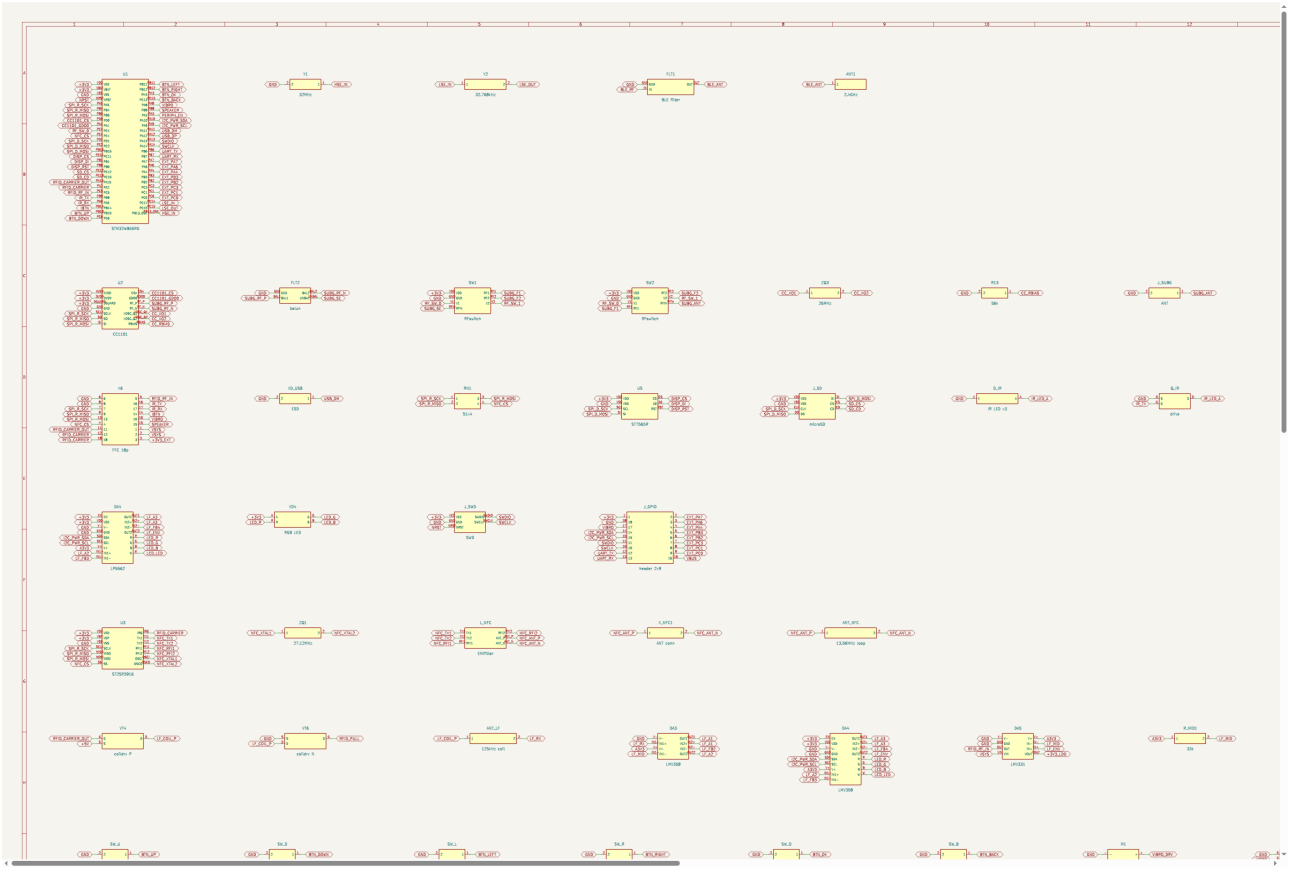
Estimated small-quantity component cost (USD). Prices are distributor ballpark figures for budgeting; get a live quote for production. Battery pack is external and listed separately. Top cost drivers shown; full costed BOM in `03_fab/NOVA-RF1_BOM_costed.csv`.

Value	MPN	Qty	Unit $	Ext $
STM32WB55RG	STM32WB55RGV6	1	6.50	6.50
ST7565R	ST7565R	1	3.80	3.80
ST25R3916	ST25R3916-AQWT	1	3.60	3.60
CC1101	CC1101RGPR	1	2.80	2.80
BQ25896	BQ25896	1	2.20	2.20
BQ27220	BQ27220	1	1.80	1.80
LM3281	LM3281YFQR	2	0.90	1.80
RFswitch	BGS13S4N9	2	0.65	1.30
LP5562	LP5562TMX	1	1.20	1.20
FFC 18p	FH34SRJ-18S-0.5SH	1	0.70	0.70
balun	B0310J50100AHF	1	0.55	0.55
microSD	DM3D-SF	1	0.55	0.55
IR RX	TSOP75338TR	1	0.50	0.50
vibro	coin ERM	1	0.50	0.50
BLE filter	LFL182G45TC1A108	1	0.45	0.45
speaker	BCE-MX8530A-03627	1	0.45	0.45
USB-C	TYPE-C-31-M-12	1	0.45	0.45
LMV358	LMV358AIDGKR	2	0.20	0.40
loadsw	TPS22917	1	0.40	0.40
32MHz	ABM8-32.000MHZ	1	0.30	0.30
2.4GHz	1001312	1	0.30	0.30
ANT	U.FL-R-SMT-1	1	0.30	0.30
...	+ 41 more lines			

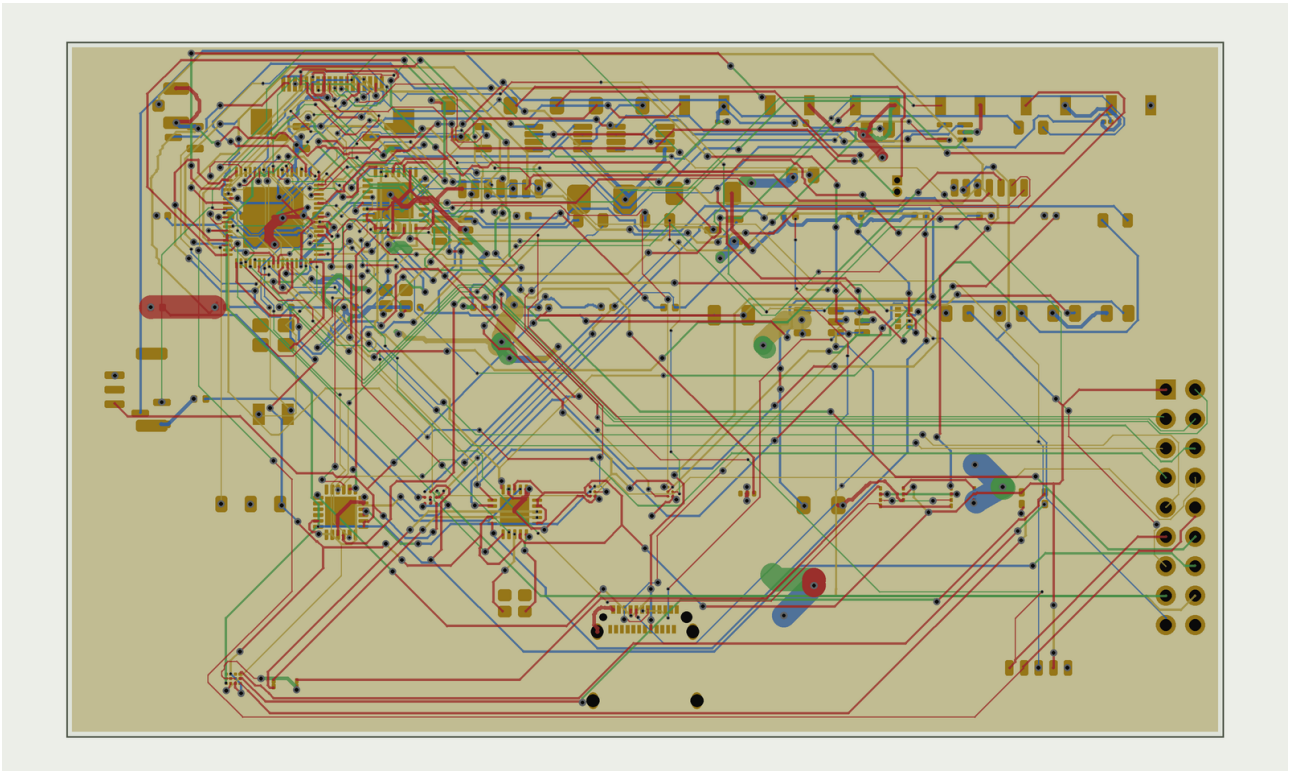
Cost line (per unit, low volume)	USD
Components (86 placements)	34.75
4-layer PCB (100x60, HDI, ~5-10 pcs)	~10.00
SMT assembly (est.)	~8.00
Li-Po battery (external)	~3.00
Enclosure (3D print, est.)	~2.00
ESTIMATED TOTAL / unit	**~57.75**

At small volume NOVA-RF1 lands around $55-60 built. Components alone are ~$35; the STM32WB55, ST25R3916 and CC1101 are the biggest drivers. At production volume (1k+) component cost roughly halves and PCB+assembly drop sharply.

3. PCB & enclosure



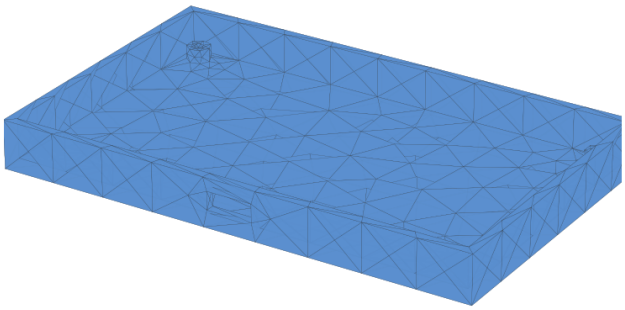
Full schematic (88 parts, 108 nets) - ERC passes with 0 errors.

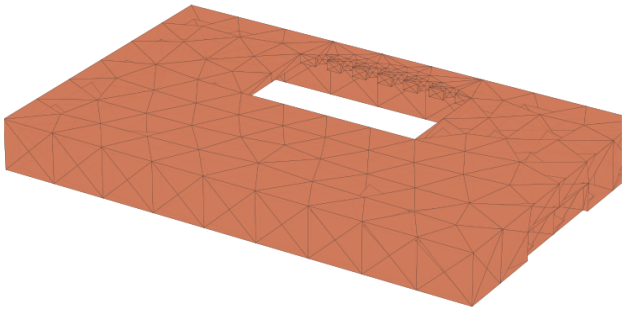


4-layer autorouted board (red=top, blue=bottom, inner=GND/PWR, dots=vias) with poured GND plane.

Bottom tray (holds PCB)

Top lid (display + buttons)





Two-part handheld enclosure: tray (PCB + USB-C cutout) and lid (display window, button holes, GPIO slot).

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Generated from the open design in projects/12_multitool-rf-board. Radios need type approval before sale.